INTRODUCTION TO FOURIER ANALYSIS ON GROUPS: HAAR MEASURE, PONTRYAGIN DUALITY, & METHODS

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Abstract. In this paper, we will give an expository overview of how to build the tools necessary to do Fourier Analysis over Locally Compact Abelian Groups. Particular emphasis will be put on making sure the basic definitions have some intuitive justification.

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1. Fourier Analysis over $\mathbb{R}^n$

Definition 1.0.1 (Fourier Series)**.** [SS03][pg. 34] The Fourier series is defined on the interval $[a, b]$.

$$\sum_{n=-\infty}^{\infty} \hat{f}(n)e^{2\pi ixn/L}, \hat{f}(n) = \frac{1}{L}\int_a^b f(x)e^{-2\pi inx/L}dx, L = b - a$$

It is valuable to note that in the standard topology of $\mathbb{R}^n$, $[a, b]$ is a compact interval. This notion of compactness will be a recurring theme as we approach Locally Compact Abelian Groups, though in this particular case it has little relevance for our purposes, since for the remainder of the paper we will not be directly dealing with Fourier Series.

Definition 1.0.2 (Fourier Transform)**.** [SS03][pg. 130] The Transform is defined as follows. ‘

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)e^{-2\pi ix\xi}dx$$

As we know, by Euler's Identity:

$$e^{-2\pi ix\xi} = \cos(-2\pi x\xi) + \sin(-2\pi x\xi)i$$

This integral will thus give us a single complex value which measures in some sense the relative strength of the frequency ξ in the function $f(x)$. We can re-construct $f(x)$ by integrating over all ξ with the following formula:

$$f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi xi\xi}d\xi$$

Now, we want to ask a somewhat unusual, albeit natural question for traditional real analysis. Can we construct similar constructs on different algebraic structures? Intuitively, a good candidate might of course be structures such as fields. However, this slightly weak. Monoids on the other hands, are likely too general to support such a structure. Further, we need to have slightly more, in particular topological properties, since we implicitly assume topological properties of $\mathbb{R}$ for Fourier Analysis. What we will see in this paper is that we will be able to support the notation of the Fourier Transform, along with Fourier Inversion over "Locally Compact Abelian Groups". In particular, we will focus on moving rather fast to build up how various disparate ideas come together to build the theory of Fourier Analysis on Groups.

2. Algebraic & Topological Properties

As previously mentioned, we need some topological and algebraic properties, its rather straight forward to just provide them. Before providing these definitions though, it is unfortunately the case that justifying the definitions, in the sense that we comprehensively build up why we use these definitions over other definitions is sufficiently out of scope for this paper that we will omit it.

Definition 2.0.1 (Homomorphism of Groups)**.** If we have two groups $(A, \star), (B, \circ)$, then χ is a homomorphism between them if the following relation holds.

$$\chi(a \star b) = \chi(a) \circ \chi(b)$$

For a newer reader, it is worth reflecting on how this definition captures the structural algebraic similarities between A under the operation $\star$ with B under the operation $\circ$ before continuing to the next definition.

Definition 2.0.2 (Topological Group)**.** [DE09][pg. 1] A group, $(G, \star)$ is a topological group when equipped with a topology τ if both hold:

(1) The map $\star$ is topologically continuous. (That is if $V \in \tau$, then $\star^{-1}(V) \in \tau_{G\times G}$, where $\tau_{G\times G}$ is the product topology)
(2) The inversion map: $I : x \mapsto x^{-1}$ is also topologically continuous (If $V \in \tau$, then $I^{-1}(V) \in \tau$).

Definition 2.0.3 (Locally Compact Group)**.** [DE09][pg. 6] A Topological Group is a Locally Compact Group if the following simultaneously holds:

(1) The Topological Group is Hausdorff. (That is $\forall x, y \in X$ when $x \neq y$, $\exists U_1, U_2 \in \tau$, so that $x \in U_1$, $y \in U_2$, and $U_1 \cap U_2 = \emptyset$)
(2) The Topological Group is Locally Compact. (That is $\forall x \in X, \exists S \in \tau$, with $x \in S$, where the closure of S, given by $\bar{S}$ is compact)
(3) (Optional) If $\star$ is abelian, that is for all $a, b \in G$, we have that $a \star b = b \star a$, then we have a *Locally Compact Abelian Group* also known as an *LCA Group*.

Example 1. $(\mathbb{R}, +)$ Equipped with the standard topology is a locally compact abelian (topological) group.

Now we have the basic topological tools, we will need to build up the tools for 'proper' Fourier Analysis. Before continuing, at a high-level, our goal will be construct analogues to typical Fourier Analysis, which reduce conceptually to the well-known definitions from

Section 1. To start with, we will need the circle group for building at tool which reduces to the typical Fourier Transform case.

Definition 2.0.4 (The Circle Group)**.** We define the circle group as the set $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, together with the binary operation of standard complex value multiplication. We leave it as an exercise to see why this is a group, but it easiest to see it in the $e^{i\theta}$ form.

Definition 2.0.5 (Characters of a Locally Compact Abelian Group)**.** [DE09][pg. 29] If we let $\chi : A \to \mathbb{T}$ be a *topologically continuous* homomorphism between a locally compact abelian group denoted by A, to the circle group $\mathbb{T}$, we call χ a character of A.

Example 2 (Cont. Chars. of $\mathbb{R}$)**.** [Fol][pg. 98] A continuous character of $\mathbb{R}$ can be described by the mapping (for fixed $\xi \in \mathbb{R}$.):

$$\chi_\xi : \mathbb{R} \to \mathbb{T}$$
$$x \mapsto e^{2\pi i \xi x}$$

Remark 2.0.6. We see how this example hints to us that pursuing this avenue further we get something similar as in our typical case for the Fourier Transform on $\mathbb{R}$ which is what we are looking for. However, it is worth putting emphasis on next point now before we continue. The characters χ act as a way to *generalize* the idea of frequency in the LCA space. This conceptual idea of thinking of this as a way to create a concept of frequency in the LCA space underpins essentially everything else we will do.

Definition 2.0.7 (Pontryagin Dual)**.** [DE09] We define the Pontryagin Dual as the space $\widehat{G}$ where $\widehat{G} = \{\chi \,|\, \chi : G \to \mathbb{T}, \chi \text{ is a character}\}$.

Theorem 2.0.8 (Pontryagin Duality)**.** [DE09][Fol][pg. 111] A Pontryagin map $\delta : G \to \widehat{\widehat{G}}$ is an isomorphism of LCA-Groups. With $\delta_g(\chi) = \chi(g)$ for every $g \in G$ with the so called "weak* topology".

Remark 2.0.9. These two definitions allow us to construct a space (which we call the Pontryagin Dual), which is the analog of the frequency space. Recall for the standard Fourier Transform, how we defined:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx$$

This works for all $\xi \in \mathbb{R}$, since $f(x)$ is invariant with respect to ξ. So, what we have here, is a frequency space of $\mathbb{R}$. In our LCA case, $\widehat{G}$ fulfills a similar role. For the Pontryagin Duality Theorem, we can see that "the frequency space, of the frequency space of G" is *isomorphic* to G. However, this is in-fact even stronger than the definition may imply, since we also make note that these are all themselves LCA Groups. Thus, we know the topological properties are preserved, as are the algebraic properties, so we have a rich structural similarity we can exploit when we go to build the generalized Fourier Transform. The rough sketch of where we are now is as follows:

(1) We have a space $\widehat{\widehat{G}}$ which is a frequency space analog.
(2) The frequency space is composed of mappings which are well-behaved enough to reduce to the expected results in classical Fourier analysis.

(3) TO DO: We need a way to integrate over G, since the Fourier Transform is defined with an $\int$, ideally with some uniqueness concepts.
(4) TO DO: We need a coherent way to describe a space A where $f : G \to A$, which is compatible with our integration decisions, and which reduces to the expected Fourier transform in the typical case.

If we can achieve the above, we have successfully constructed an analog for Fourier Analysis over LCA Groups.

3. Haar Measure & General Fourier Transform

We left off needing a way to build integration techniques which worked for our purpose. Luckily, we have a the tools of *Measure Theory* to give us rather beautiful results which we can use. Like before, we will simply give these definitions and build the final results.

Definition 3.0.1 (Haar Measure)**.** [DE09] Let G be a locally compact group (need not be Abelian). There exists a unique up to constant non-zero left invariant radon measure on G, which we call the *Haar Measure*.

Remark 3.0.2 (Why the Haar Measure?)**.** A measure is just a way for us to abstractly generalize concepts such as lengths, areas, volumes, etc over more general sets. A borel measure is a measure which is commonly used with topological spaces. A Radon measure is a Borel measure that is finite on compact sets, outer regular on all Borel sets, and inner regular on open sets. [DM] These regularity conditions ensure that the measure is well-behaved from a topological perspective. Finally, the Fourier Transform has certain invariant properties which we want to replicate. For that reason, we put these together and look at the Haar Measure. However, the definition (which implicitly cites a theorem), says there is a *unique* Haar Measure. This is convenient. Unsurprisingly, for $\mathbb{R}$, the Haar Measure is the Lebesgue Measure, which itself is the Measure we typically use for Fourier Analysis over $\mathbb{R}$.

Remark 3.0.3 (Measure Theoretic Integral)**.** [Tao][Ch. 1.4] There is a deep theory for measure theoretic integration which we will omit for brevity. However, the key results we need are that we can construct integrals over measure spaces with respect the function and the measure μ and get integration results which are generalizations of our well known integration results from basic analysis. This takes the notation of:

$$\int_X f \, d\mu$$

Call f *μ-integrable* ($f \in L^1(\mu)$, that is integrable with respect to the measure) if both are finite.

Definition 3.0.4 (Haar Integral)**.** The Haar Integral is the Measure Theoretic Integral corresponding to the Haar Measure, similar to the Lebesgue Integral corresponding to the Lebesgue Measure.

With this, we have checked off our 3rd item on the list at the end of section two. Now, we finally have enough depth to construct our Fourier Transform, since we have a coherent way to complete integration over the space.

4. Conclusion for Fourier Transform over LCA Groups

Now we have nearly all of the pieces we need to construct the Fourier Theory over LCA Groups, so lets put together these pieces. Let us define $f : G \to \mathbb{C}$, where G is an LCA Group. Recall, that when we take $\chi(g).g \in G$, that $\chi(g) \in \mathbb{T}$. This the product $f(g) \cdot \chi(g) \in \mathbb{C}$, with $g \in G$. So, for this theory, we will restrict ourselves to functions f which map from G to the complex numbers, which is a notable constraint. Since we have the goal that this reduces to the typical Fourier Transform, we will take the conjugate of $\chi(g)$, since it is complex, (denoted by $\overline{\chi(g)}$. Lastly, we integrate over our Haar Measure, and we have a notion of a Fourier Transform over G.

Definition 4.0.1 (Fourier Transform Over an LCA Group)**.** [DE09] The Fourier transform is the function $\widehat{f}$ on $\widehat{G}$ defined by

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}d\mu(x)$$

This is a nice construction, but still lack a theory of Fourier Inversion, so it is hard to know that this is complete. However, we have not yet used one of our tools, which is the Pontryagin Duality Theorem. By this theorem, we can conclude that $\exists \widehat{\widehat{f}} : \widehat{\widehat{G}} \to \mathbb{C}$:

$$\widehat{\widehat{f}} \cong f$$

We want our inversion process to associate this $\widehat{\widehat{f}}$ with f. This is not enough to prove inversion, which requires extra work, but, it is a component of the idea. The inversion theorem given below.

Theorem 4.0.2 (Fourier Inversion Formula)**.** [Fol][pg. 111] If $f \in L^1(G)$ and $\widehat{f} \in L^1(\widehat{G})$ we have, almost everywhere.

$$f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x)\widehat{\mu}(\chi)$$

If f is continuous, then for all x.

So now, in total, we have a comprehensive overview of the two core tools for Fourier Analysis over Groups are constructed. We can go much further and expand upon this with specific results. We can also start to ask questions like, what if we weaken certain conditions. For instance, what if the groups are non-abelian? In-fact, we also have some theory for these classes of groups as well, but we will conclude here.

References

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